

Sahil Kumar

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Professional Summary

Full Stack Software Engineer with experience building scalable web applications, real-time systems, and AI-powered products. Skilled in React, Next.js, Node.js, TypeScript, PostgreSQL, Redis, and cloud-native architectures, with a strong focus on delivering performant end-to-end user experiences and production-ready systems.

Technical Skills

Full Stack Development: React.js, Next.js, Node.js, Express.js, REST APIs, WebSockets, Authentication (JWT), Authorization, HTML, CSS, TailwindCSS

Databases & Infrastructure: PostgreSQL, MongoDB, Redis, Docker, AWS, Linux, Git, GitHub

AI / Generative AI: LLM Application Development, Prompt Engineering, RAG Pipelines, LangChain, Vector Databases (Pinecone, FAISS), Semantic Search, Embedding Pipelines

Programming Languages: C++, Python, JavaScript, TypeScript, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Experience

Software Development Intern — Rawyal

Jan-Apr 2026

Node.js, Express.js, Python, Redis, BullMQ, Pinecone, MongoDB, RAG

- Built a scalable asynchronous document processing pipeline supporting PDF, Office, HTML, and image transformations, reducing processing latency by 35%.
- Developed a distributed BullMQ-based job system with retries, concurrency control, and failure isolation, improving throughput by 40% under parallel workloads.
- Engineered a RAG-based retrieval system using vector embeddings and Pinecone, enabling context-aware search and querying across large document datasets.

Software Developer Intern — ModelSuite.ai

Sept-Dec 2025

React.js, Node.js, Express.js, MongoDB, Zustand

- Developed scalable MERN-stack features, reusable React components, and modular architecture of the platform.
- Improved application performance by reducing load times by 40% through frontend optimization, efficient API integration, and database query improvements.
- Built reusable React components and state management workflows using Zustand while collaborating in an Agile environment to deliver production-ready features.

Projects

AI-Powered Digital Twin Platform

[GitHub](#) | [Live](#)

Next.js, TypeScript, Node.js, FastAPI, PostgreSQL, Redis, Socket.io, RAG, Pinecone

- Built a full-stack digital twin platform for real-time simulation, monitoring, and AI-powered system analysis.
- Designed a simulation engine supporting 100+ concurrent component simulations and generation of high-frequency telemetry data.
- Engineered a Redis- and Socket.io-based real-time data pipeline processing 15k+ telemetry events per minute for live dashboards, alerts, and analytics.
- Integrated AI-powered anomaly detection, failure prediction, and semantic search using FastAPI microservices, Pinecone, and LLM-based reasoning.
- Architected scalable backend services for telemetry ingestion, event processing, and real-time analytics.

LifeFlow – Real-Time Blood Donation Platform

[GitHub](#) | [Live](#)

React.js, Node.js, Express.js, MongoDB, Redis, BullMQ, JWT

- Built a full-stack blood donation platform for real-time donor matching and emergency blood request management.
- Developed responsive React dashboards and implemented JWT-based authentication with role-based access control (RBAC), supporting secure workflows for 100+ donor accounts.
- Designed 10+ RESTful APIs and engineered geolocation-based donor matching using MongoDB geospatial indexing and Redis caching, achieving sub-10ms nearby donor discovery.
- Reduced API latency by 15–30× and built fault-tolerant asynchronous workflows using Redis and BullMQ for reliable SMS and notification delivery.

Education

Indian Institute of Information Technology, Una

2023 – Present

B.Tech in Electronics and Communication Engineering

CGPA: 7.04

Achievements

- Specialist (Codeforces), 3 Star (CodeChef), 300+ problems solved, Global Rank 2200 in LeetCode Weekly Contest.